

Anonymization of the Adult Dataset with Risk and Utility Analysis

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1 Attribute Classification

After preprocessing, all attributes were classified according to their role in privacy protection.

Summary:

- **Quasi-identifiers (10):** age, workclass, education, marital-status, occupation, relationship, race, sex, hours-per-week, native-country.
- **Sensitive (3):** capital-gain, capital-loss, income.
- **Insensitive (2):** fnlwgt, education-num.
- **Identifying:** None.

Table 1: Attribute Classification with Justifications

Attribute	Type	Justification (Direct & Abbreviated)
age	QID	Helps identify people when combined with occupation, marital status, or country.
workclass	QID	Employment sector. Weak alone, but useful with occupation and age.
fnlwgt	Insensitive	Survey weight. Too many unique values; excluding preserves utility.
education	QID	Education level. Strong differentiator for attackers.
education-num	Insensitive	Same as education (numeric). Redundant; excluded to avoid over-generalization.
marital-status	QID	Marital status. Narrows down individuals with age and relationship.
occupation	QID	Job role. High re-identification power.
relationship	QID	Family role. Adds extra distinction beyond marital status.
race	QID	Racial category. Reduces possible individual set.
sex	QID	Gender. Weak alone, strong in combinations.
capital-gain	Sensitive	Private investment income. Protect from disclosure.
capital-loss	Sensitive	Private investment losses. Protect from disclosure.
hours-per-week	QID	Work hours. Distinguishes part-time vs full-time.
native-country	QID	Country of origin. Many rare values increase re-ID risk.
income	Sensitive	Salary class ($>50K/\leq 50K$). Primary attribute to protect.

2 Re-identification Risk of the Original Dataset

The original dataset was analyzed using the ARX Risk Analysis module.

Measure	Value [%]
Lowest prosecutor risk	4.34783%
Records affected by lowest risk	0.07064%
Average prosecutor risk	84.50293%
Highest prosecutor risk	100%
Records affected by highest risk	76.17088%
Estimated prosecutor risk	100%
Estimated journalist risk	100%
Estimated marketer risk	84.50293%
Sample uniques	76.17088%
Population uniques	16.96566%

Figure 1: Risk

- Average prosecutor risk: 84.5%
- Highest prosecutor risk: 100%
- Sample uniqueness: 76.2%

The original dataset exhibits a high re-identification risk because many combinations of quasi-identifiers are unique.

3 Anonymization Models and Parameter Analysis

3.1 Hierarchy Design

Generalization hierarchies were created for all quasi-identifiers.

Table 2: Hierarchy Design Summary for Quasi-Identifiers

Attribute	Hierarchy Type	Structure	Rationale
age	Interval-based	5-year intervals $\rightarrow$ [17, 22[, ..., [92, 97[	Numeric, high cardinality; preserves order while reducing precision.
education	Order-based	4×4 (16 $\rightarrow$ 4 $\rightarrow$ 4)	Ordinal values; semantic grouping into educational levels.
occupation	Order-based	5×3 (15 $\rightarrow$ 5 $\rightarrow$ 3)	Categorical with natural job sectors; manual grouping preserves meaning.
native-country	Redaction-based	Character masking (right to left)	Many rare values; masking avoids manual grouping and protects small groups.

3.2 k-Anonymity Analysis

Table 3: k-Anonymity (0% suppression)

k	Avg. Risk (%)	Highest Risk (%)	Uniqueness (%)	Info Loss (%)
Original	84.5	100.0	76.2	0.0
5	2.84	20.00	0.0	49.2
10	1.64	10.00	0.0	53.9
15	1.20	6.67	0.0	56.9
20	0.95	5.00	0.0	59.2
25	0.81	4.00	0.0	60.8

For k -anonymity, the sensitive attributes **income**, **capital-gain**, and **capital-loss** were reclassified as *insensitive*. This allows the anonymization process to focus generalization on the quasi-identifiers only, which is standard practice for k -anonymity since it does not protect against attribute disclosure.

The original dataset had an average prosecutor risk of 84.5% and sample uniqueness of 76.2%, confirming high vulnerability to linkage attacks. Applying k -anonymity (0% suppression) reduced risk dramatically: at $k = 5$, average risk dropped to 2.84% and uniqueness to 0%. As k increased, risk continued to fall monotonically, reaching 0.81% at $k = 25$.

Utility decreased with higher k : the utility score (where lower is better, according to the Loss metric in ARX) increased from 0.377 at $k = 5$ to 0.466 at $k = 25$, indicating a loss of information. The trade-off shows diminishing returns: from $k = 5$ to $k = 10$, risk fell by 1.20 percentage points (42% relative reduction) while the utility score increased by 0.036; from $k = 20$ to $k = 25$, risk fell by only 0.14 percentage points (15% relative reduction) while the utility score increased by 0.013. Therefore, $k = 10$ or $k = 15$ offers a practical balance, where risk is already below 2% and further privacy gains are marginal relative to the utility cost.

Increasing k reduces re-identification risk because individuals become indistinguishable from larger groups. However, larger values require stronger generalization and therefore reduce data utility.

3.3 l-Diversity Analysis

For l -diversity, k -anonymity was fixed at $k = 10$. The sensitive attribute **income** was protected using distinct l -diversity. Two suppression limits were tested: 0% (no record removal) and 100% (allowing arbitrary suppression).

Table 4: l -Diversity Results ($k = 10$, varying suppression)

l	Suppression Limit	Avg. Risk (%)	Highest Risk (%)	Info Loss (%)
2	0%	0.96	10.0	64.4
2	100%	0.96	10.0	0.0
3	0%	0.00	0.0	100.0
3	100%	0.00	0.0	0
4	0%	0.00	0.0	100.0
4	100%	0.00	0.0	0

For $l = 2$, both suppression limits achieve nearly identical low risk (0.96% average, 10% highest). However, the utility differs dramatically: with 0% suppression, information loss is 64.4% (heavy generalisation); with 100% suppression, information loss is 0% because ARX instead suppresses a small number of problematic records to satisfy the constraint. This demonstrates the classic trade-off: suppression preserves data detail for the remaining records, while generalisation preserves all records but reduces precision.

For $l = 3$ and $l = 4$, the dataset cannot support the diversity constraint under any suppression limit. At 0% suppression, information loss reaches 100% (generalisation destroys all utility); at 100% suppression, all records are suppressed, leaving an empty output. This is due to the highly imbalanced distribution of **income** (approx. 75% $\leq$ 50K, 25% $>$ 50K). Consequently, $l = 2$ is the only practical configuration, and a small suppression limit (e.g., 5–10%) would likely achieve the best balance: low risk, moderate utility, and minimal information loss from generalisation.

3.4 Suppression Limit Analysis

The effect of the suppression limit was evaluated for l -diversity. Table 5 compares two extreme limits: 0% (no record removal, only generalisation) and 100% (any number of records may be removed).

Table 5: Effect of suppression limit on l -diversity ($k = 10, l = 2$)

Suppression Limit	Avg. Risk (%)	Info Loss (%)	Observation
0%	0.96	64.4	Heavy generalisation, all records kept
100%	0.96	0.0	No generalisation, few records suppressed

With a 100% suppression limit, ARX achieved the same low risk (0.96%) but with zero information loss, because it removed a small number of problematic records instead of generalising all quasi-identifiers. This illustrates the classic trade-off: suppression preserves data detail for the remaining records, while generalisation preserves all records but reduces precision. In practice, a moderate suppression limit (e.g., 5–10%) would likely provide the best balance: low risk, moderate utility, and only a tiny fraction of records removed.

3.5 Comparison of Privacy Models

Table 6 summarises the best configurations obtained for each privacy model under 0% suppression (to allow a fair comparison).

Table 6: Comparison of best privacy models (0% suppression)

Model	Avg. Risk (%)	Info Loss (%)	Notes
Original dataset	84.5	0.0	High re-identification risk
k -Anonymity ($k = 10$)	1.64	53.9	Good balance, no suppression
l -Diversity ($l = 2, k = 10$)	0.96	64.4	Lower risk, higher information loss

Both anonymisation models drastically reduce re-identification risk compared to the original data. k -Anonymity achieves a slightly higher risk (1.64%) but lower information loss (53.9%), while l -Diversity further lowers risk to 0.96% at the cost of additional information loss (64.4%).

4 Conclusion

This study demonstrated the use of ARX to anonymise the Adult dataset and evaluate privacy risks.

- The original dataset exhibited a very high re-identification risk (average prosecutor risk 84.5%, sample uniqueness 76.2%) because many combinations of quasi-identifiers were unique. Applying k -anonymity with $k = 10$ reduced the average risk to 1.64% and eliminated uniqueness, but introduced an information loss of 53.9% due to generalisation. Adding l -diversity ($l = 2$) on the `income` attribute further lowered risk to 0.96% at the cost of raising information loss to 64.4%.
- The results clearly illustrate the privacy–utility trade-off inherent in data anonymisation. Stronger privacy guarantees (higher k , l -diversity) generally require increased information loss or record suppression.

AI Use Declaration

AI used to assist with report structure, LaTeX formatting, and preparation of tables.

References

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